THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

Software Testing

LAB TASK 2



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1. Application Under Test

Item	Details
Application	TodoMatic (todo-react), a sample to-do list built with React
GitHub repository	https://github.com/mdn/todo-react
Branch / commit	main / d28218cfb353d474c0cb51b495325db6eb908622 (committed Oct 19, 2026)
Application version	package.json version 0.3.0 (React 18.2.0, Vite 5.4.21)
How it was run	Node.js v24 (see .nvmrc), Yarn. yarn then yarn dev (or npm install, npm run dev), then open http://localhost:3000. Note: the README says yarn start, but package.json defines only dev, build, lint and preview.
Same application for Lab 03	Yes. The same application, rules and test cases will be used in Lab Task 03.

2. Users and Main Features

Users: one type of user, the end user, who manages a personal list of tasks in the browser (see A-1).

ID	Feature	Rules covering it
F-1	View the task list and the count heading	R-01, R-09
F-2	Add a new task	R-02, R-03
F-3	Mark a task completed or active	R-04
F-4	Filter tasks (All / Active / Completed)	R-05
F-5	Rename a task (Edit, Save, Cancel)	R-06, R-07
F-6	Delete a task	R-08

3. Assumptions

ID	Assumption
A-1	There is one type of user (an end user managing personal tasks). There is no login, no administrator and no server; the whole app runs in the browser.
A-2	Tasks are held in memory only. Reloading the page restores the three built-in tasks (stated for completeness; not tested in this lab).
A-3	All cases are run in a desktop browser (for example latest Chrome) at http://localhost:3000, and each case starts from a fresh page reload so that cases do not depend on each other.
A-4	"Blank" means an empty string or a string made only of space characters.
A-5	In this commit the edit field opens empty (it is not pre-filled with the old name), so edit tests type the complete new name.
A-6	The count heading follows the tasks displayed in the selected filter (R-09), not a global count of active tasks.
A-7	Rules describe the behaviour the application is required to have. Where the running application differs from a rule, the test case will be recorded as failed when executed in Lab Task 03.

4. Application Rules (Acceptance Criteria)

Rule ID	Name	Required behaviour
R-01	Initial state	When the application loads, the list contains exactly three tasks in this order: Eat (completed), Sleep (active), Repeat (active). The All filter is selected and the new-task input is empty.
R-02	Add a task	Entering a non-blank name in the new-task input and clicking Add appends one new task to the end of the list, as active (unchecked), with Edit and Delete buttons. The input is cleared. Existing tasks are unchanged.
R-03	Reject blank new task	If the new-task input is empty or contains only spaces, clicking Add must not create a task. The task list and the count heading stay as they were.
R-04	Toggle completion	Ticking a task's checkbox marks that task completed; unticking marks it active again. Only that task changes.
R-05	Filters	The buttons All, Active and Completed are available and exactly one is pressed at a time. All lists every task; Active lists only tasks that are not completed; Completed lists only completed tasks. Switching filters never changes task data.
R-06	Rename a task	Clicking Edit shows a text field with Cancel and Save. Saving a non-blank name replaces that task's name and keeps its position and completed status. Clicking Cancel discards the typed text and keeps the old name.
R-07	Reject blank rename	Saving an edit with an empty or spaces-only name must not rename the task; the original name is kept.
R-08	Delete a task	Clicking Delete removes that task permanently from every filter view. Other tasks are unchanged and keep their order.
R-09	Count heading	The heading above the list reads "N tasks remaining", where N is the number of tasks shown in the currently selected filter. The word is "task" when N is 1 and "tasks" when N is 0 or 2 or more.

5. Test Case Summary and Rule Mapping

Test case	Title	Rules linked	Type
TC-01	Application loads with the built-in tasks	R-01, R-09	Valid
TC-02	Add a valid task	R-02, R-09	Valid
TC-03	Reject an empty new task	R-03	Invalid input
TC-04	Reject a spaces-only new task	R-03, R-09	Invalid input
TC-05	Mark a task completed and view it in the Active filter	R-04, R-05, R-09	Valid
TC-06	Mark a task active again and check an empty Completed filter	R-04, R-05, R-09	Valid / boundary
TC-07	Rename a task with a valid name	R-06, R-09	Valid
TC-08	Reject renaming a task to an empty name	R-07	Invalid input
TC-09	Delete a task	R-08, R-05, R-09	Valid
TC-10	Cancel an edit	R-06	Valid (cancel action)

Rule	Covered by
R-01	TC-01
R-02	TC-02
R-03	TC-03, TC-04
R-04	TC-05, TC-06
R-05	TC-05, TC-06, TC-09
R-06	TC-07, TC-10
R-07	TC-08
R-08	TC-09
R-09	TC-01, TC-02, TC-04, TC-05, TC-06, TC-07, TC-09

6. Test Cases

Every case starts from a fresh page reload (A-3). Actual result is left blank and Status is Not run until Lab Task 03.

TC-01 - Application loads with the built-in tasks

Test-case field	TC-01 - Application loads with the built-in tasks
Title	Application loads with the built-in tasks
Requirement	R-01, R-09 Initial list, selected filter and count heading are correct at start-up.
Preconditions	Application is running at http://localhost:3000. Page has just been loaded; nothing has been clicked.
Test data	Built-in tasks: Eat (completed), Sleep, Repeat
Steps	1. Open http://localhost:3000 in a new browser tab. 2. Without clicking anything, inspect the heading, the three filter buttons, the input field and the list.
Expected result	The page title text is "TodoMatic". The All button is pressed (highlighted); Active and Completed are not. The list shows three items in this order: Eat (checkbox ticked), Sleep (unticked), Repeat (unticked), each with Edit and Delete buttons. The count heading reads "3 tasks remaining". The new-task input is empty.
Actual result	
Status	Not run

TC-02 - Add a valid task

Test-case field	TC-02 - Add a valid task
Title	Add a valid task
Requirement	R-02, R-09 A non-blank name creates one active task at the end of the list and clears the input.
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Submit Lab 02 report
Steps	1. Click the "What needs to be done?" input. 2. Type Submit Lab 02 report. 3. Click Add. 4. Inspect the list, the count heading and the input field.
Expected result	"Submit Lab 02 report" appears exactly once, as the 4th item below Repeat, with an unticked checkbox and Edit and Delete buttons. Eat, Sleep and Repeat are unchanged. The count heading reads "4 tasks remaining". The input field is empty.
Actual result	
Status	Not run

TC-03 - Reject an empty new task

Test-case field	TC-03 - Reject an empty new task
Title	Reject an empty new task
Requirement	R-03 Clicking Add with an empty input must not create a task.
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present. The new-task input is empty.
Test data	(none - input left empty)
Steps	1. Confirm the new-task input contains no text. 2. Click Add. 3. Inspect the list and the count heading.
Expected result	No new item is added; the list still shows only Eat, Sleep and Repeat. The count heading still reads "3 tasks remaining".
Actual result	
Status	Not run

TC-04 - Reject a spaces-only new task

Test-case field	TC-04 - Reject a spaces-only new task
Title	Reject a spaces-only new task
Requirement	R-03, R-09 A name made only of spaces is blank and must not create a task.
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Three space characters (" ")
Steps	1. Click the new-task input. 2. Press the space bar three times (no other characters). 3. Click Add. 4. Inspect the list and the count heading.
Expected result	No new item (and no item with a blank name) is added; the list shows only Eat, Sleep and Repeat. The count heading still reads "3 tasks remaining".
Actual result	
Status	Not run

TC-05 - Mark a task completed and view it in the Active filter

Test-case field	TC-05 - Mark a task completed and view it in the Active filter
Title	Mark a task completed and view it in the Active filter
Requirement	R-04, R-05, R-09 Ticking a task completes it; the Active filter then hides it; singular wording for one task.
Preconditions	Application is running; page freshly loaded. All view is selected; Sleep is unticked.
Test data	Task: Sleep Filter: Active
Steps	1. Tick the checkbox next to Sleep. 2. Inspect the list in the All view. 3. Click the Active filter button. 4. Inspect the list, the pressed filter button and the count heading.
Expected result	After step 1, the Sleep checkbox is ticked; Eat is still ticked; Repeat is still unticked. After step 3, only Repeat is listed (Eat and Sleep are not shown). The Active button is pressed. The count heading reads "1 task remaining".
Actual result	
Status	Not run

TC-06 - Mark a task active again and check an empty Completed filter

Test-case field	TC-06 - Mark a task active again and check an empty Completed filter
Title	Mark a task active again and check an empty Completed filter
Requirement	R-04, R-05, R-09 Unticking returns a task to active; Completed then shows nothing; zero uses the plural word.
Preconditions	Application is running; page freshly loaded. All view is selected; Eat is ticked (completed).
Test data	Task: Eat Filters: Completed, then Active
Steps	1. Untick the checkbox next to Eat. 2. Click the Completed filter button and inspect the list and heading. 3. Click the Active filter button and inspect the list and heading.
Expected result	After step 1, the Eat checkbox is unticked. After step 2, the list is empty, the Completed button is pressed and the heading reads "0 tasks remaining". After step 3, Eat, Sleep and Repeat are all listed, the Active button is pressed and the heading reads "3 tasks remaining".
Actual result	
Status	Not run

TC-07 - Rename a task with a valid name

Test-case field	TC-07 - Rename a task with a valid name
Title	Rename a task with a valid name
Requirement	R-06, R-09 Saving a non-blank name replaces the task name and keeps its position and status.
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Task to edit: Sleep New name: Sleep 8 hours
Steps	1. Click Edit next to Sleep. 2. Confirm a text field with Cancel and Save buttons is shown for that task. 3. Click the text field and type Sleep 8 hours. 4. Click Save. 5. Inspect the list and the count heading.
Expected result	After step 2, the edit form (text field, Cancel, Save) replaces the normal view of the Sleep task only. After step 4, the edit form closes and the second item reads "Sleep 8 hours" with an unticked checkbox. No item named "Sleep" remains; Eat (ticked) is first and Repeat is third. The count heading reads "3 tasks remaining".
Actual result	
Status	Not run

TC-08 - Reject renaming a task to an empty name

Test-case field	TC-08 - Reject renaming a task to an empty name
Title	Reject renaming a task to an empty name
Requirement	R-07 Saving an empty name must not rename the task.
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Task to edit: Repeat New name: (none - field left empty)
Steps	1. Click Edit next to Repeat. 2. Leave the text field empty. 3. Click Save. 4. Inspect the list.
Expected result	The task is still named "Repeat"; no item with a blank name is shown. The list shows Eat, Sleep, Repeat in that order and the count heading reads "3 tasks remaining".
Actual result	
Status	Not run

TC-09 - Delete a task

Test-case field	TC-09 - Delete a task
Title	Delete a task
Requirement	R-08, R-05, R-09 Delete removes the task from every view and leaves the others unchanged.
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Task to delete: Sleep
Steps	1. Click Delete next to Sleep. 2. Inspect the list and the count heading. 3. Click the Active filter button and inspect the list and heading.
Expected result	After step 1, Sleep is gone; the list shows Eat (ticked) then Repeat (unticked) and the heading reads "2 tasks remaining". After step 3, only Repeat is listed (Sleep does not reappear) and the heading reads "1 task remaining".
Actual result	
Status	Not run

TC-10 - Cancel an edit

Test-case field	TC-10 - Cancel an edit
Title	Cancel an edit
Requirement	R-06 Cancel discards the typed text and keeps the old name.
Preconditions	Application is running; page freshly loaded. All view is selected; the three built-in tasks are present.
Test data	Task to edit: Repeat Typed text: Do not save
Steps	1. Click Edit next to Repeat. 2. Type Do not save in the text field. 3. Click Cancel. 4. Inspect the list.
Expected result	The edit form closes and Repeat is shown again in its normal view with an unticked checkbox. The task is still named "Repeat"; the text "Do not save" appears nowhere in the list. The list shows Eat, Sleep, Repeat and the count heading reads "3 tasks remaining".
Actual result	
Status	Not run

THANK YOU SO MUCH